CAPSTONE TWO · SPRINGBOARD DATA SCIENCE

Predicting wafer failures from process sensors

A time-ordered study of the UCI SECOM semiconductor dataset

Presenter: Thomas Truong
Mentor: Lucas Allen

The question

Can in-line sensor readings flag a wafer that will fail final test, before more cost is spent on it?

- Pass or fail is known only at final test, after most of the cost is spent
- An early flag could prioritise inspection and save rework
- The task: rank wafers by failure risk using 590 sensors per unit

The data: 1,567 units, 590 sensors, 104 failures

1,567

production units

Jul 19 to Oct 17, 2008

590

sensor readings per unit

many missing, constant or duplicated

6.6%

fail final test

about 13 passes per failure

Always predicting “pass” is 93% accurate and catches nothing. So models are judged on **PR AUC**, where a random model scores only about 0.054.

Split by time before any cleaning

Train
1,253 units · 87 failures · up to Oct 2

Test
314 units · 17 failures

- The model must score tomorrow’s wafer knowing only the past
- A random split would train on the future and overstate results
- Medians, clipping limits and scalers are learned from train only

Cleaning: 590 sensors became 464 columns

−24

mostly missing

over 50% absent

−122

constant

no information

−8

duplicated

correlation above 0.9999

+28

missing flags

a sensor going offline can itself signal a fault

Missing values filled with training medians. Extreme readings were **clipped** to the 0.5% to 99.5% range, never deleted: with only 104 failures, extreme readings are likely the failures. No units were removed.

No sensor separates failures on its own

0.728

best single-sensor AUC (S059)

- 20 of 436 sensors are significant after false-discovery correction
- 70 pass an uncorrected 0.05 test, against about 22 expected by chance
- The median sensor sits just 0.024 away from random (AUC 0.5)

Weak signal spread across correlated sensors

118

of 436 sensors are heavily skewed

so scaling happens inside the model pipeline

298

sensors have a partner above r = 0.9

blocks of redundant sensors favour regularised models

1.37×

failure rate when a sensor is missing

8.8% against 6.4%, so the flags were kept

Failures are shifted and more spread out than passes, but 72% sit inside the central range of two top sensors at once, so no simple rule works.

Three models, tuned on the past, tested on the future

Logistic Regression

L1 penalty with top-k feature selection. Simple and interpretable.

Random Forest

Bagged trees. Handles many noisy features and non-linear effects.

LightGBM

Gradient boosting. Strong on tabular data.

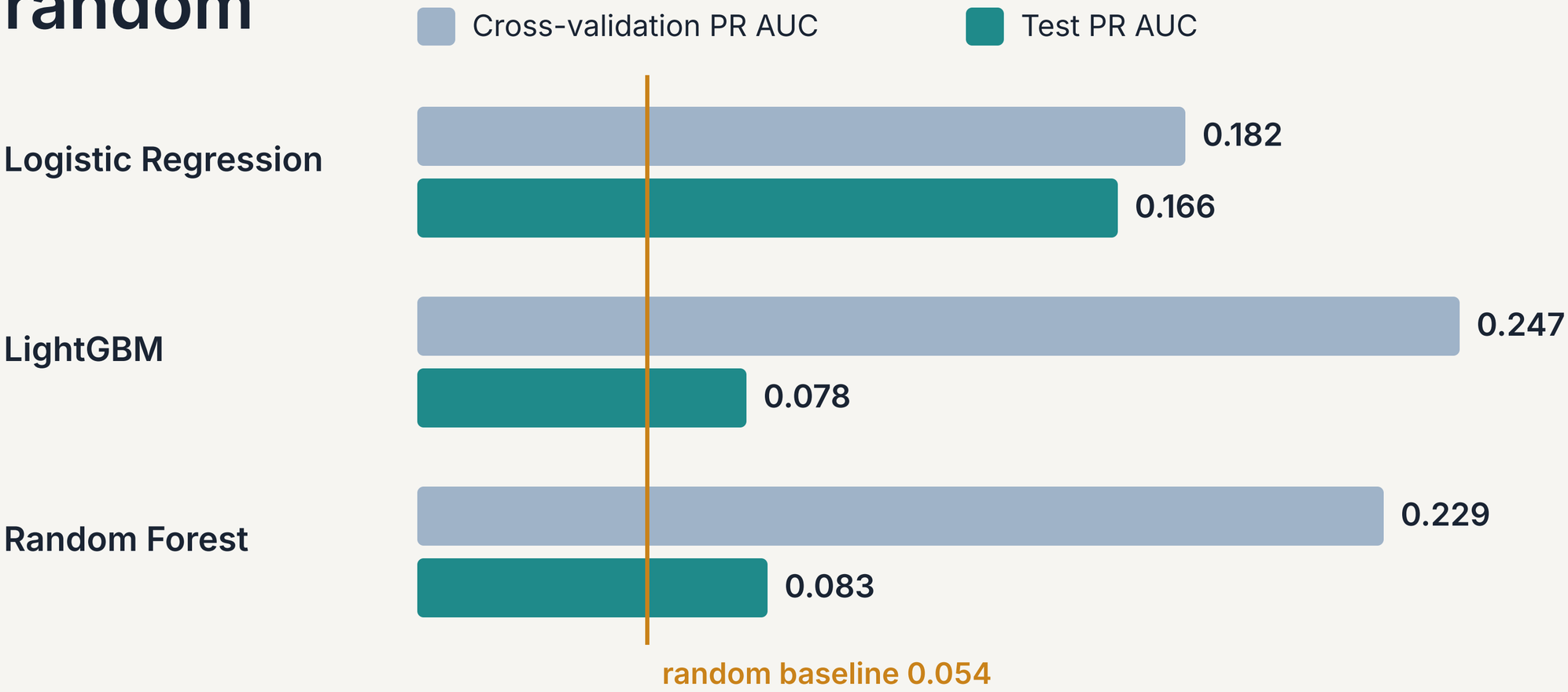
- Class weights for the 13-to-1 imbalance
- 5-fold **time-ordered** cross-validation (train on past, validate on future, gap of 10)
- Scored on PR AUC; test set touched once, after tuning

Only Logistic Regression held up on the test period

Model	CV PR AUC	Test PR AUC	Test ROC AUC	Recall	Precision	F1
Logistic Regression	**0.182**	**0.166**	**0.574**	**0.294**	**0.106**	**0.156**
LightGBM	0.247	0.078	0.580	0.000	0.000	0.000
Random Forest	0.229	0.083	0.605	0.000	0.000	0.000
Random baseline	n/a	0.054	0.500	n/a	n/a	n/a

314 test units, 17 failures, default 0.5 cutoff. Recall, precision and F1 are for the failure class.

Cross-validation misled: the tree models fell to near random



Final model: Logistic Regression caught 5 of 17 failures

5
failures caught

12
failures missed

42
false alarms

255
passes correct

- Test PR AUC **0.166**, about 3× baseline
- Recall **0.294**
- Precision **0.106**
- F1 **0.156**

What the result can and cannot support

- **Only 17 test failures.** One more or fewer catch moves recall by about 6 points.
- **Model choice rests on one test result.** Cross-validation preferred LightGBM.
- **Low precision.** About 9 in 10 alerts are false alarms.
- **Weak signal.** Test ROC AUC is only 0.57 to 0.61.
- **Three months of data.** Longer-term drift cannot be assessed.

Recommendation and next steps

Use the model as a screening aid to prioritise wafers for inspection, not as an automatic reject rule.

1. Bootstrap confidence intervals on test PR AUC, recall and precision
2. Instability features: rolling standard deviation, absolute z-score, lags
3. Walk-forward validation over many time windows
4. Calibrate probabilities and weigh missed failures against false alarms
5. Collect more failure examples

Questions and feedback

Weak but real signal. Logistic Regression: test PR AUC 0.166 against a 0.054 baseline, 5 of 17 failures caught.